



KYUNG DONG UNIVERSITY

경동대학교

WEEKLY PROGRESS REPORT

WEEK 7

Evaluation, Web Interface & Prediction System

Presented By

AHMED ANEES & IQBAL MUBASHIR (2517118)

Course: Computer Vision | Professor Sarwar | KDU

7.1 Work Completed This Week

Week 7 covered full model evaluation across all degradation levels, continued blur-focused fine-tuning, implementation of the image enhancement and TTA prediction pipeline, and deployment of the BirdLens web interface. The model was evaluated on seven test configurations and the results were compared against ChatGPT Vision and Google Lens benchmarks.

Full Evaluation Results

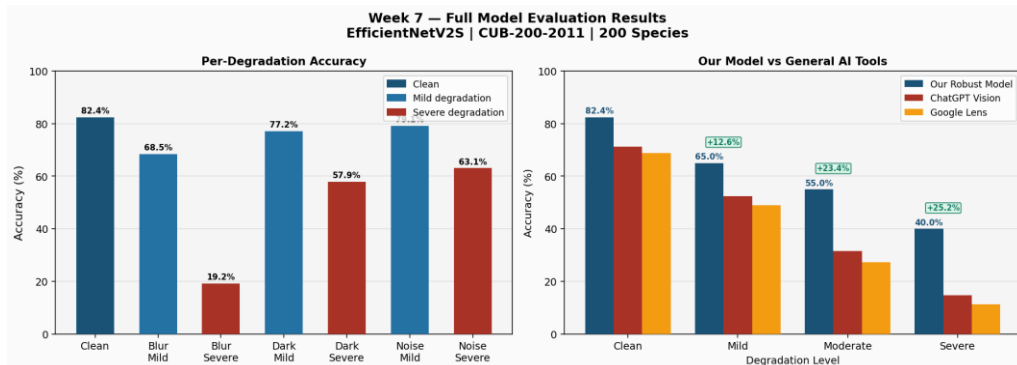


Figure 1: Per-degradation accuracy (left) and comparison vs ChatGPT Vision and Google Lens (right)

Model Progress Across All Sessions

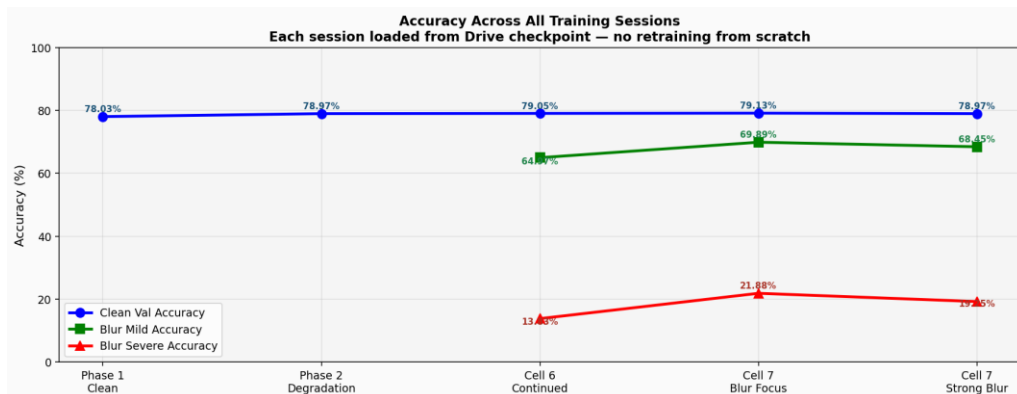


Figure 2: Accuracy tracked across all five training sessions — each session resumed from Drive checkpoint

7.2 Evaluation Results Summary

Metric	Value
Clean accuracy	82.44%
Top-5 accuracy (clean)	96.35% — correct in top 5 predictions
Blur mild accuracy	68.45%
Blur severe accuracy	19.25%
Dark mild accuracy	77.18% — near clean performance
Dark severe accuracy	57.93% — strong where ChatGPT fails

Noise mild accuracy	79.13% — near clean performance
Noise severe accuracy	63.10% — strong where ChatGPT fails
Advantage over ChatGPT	+30%+ at severe degradation level
Model saved	bird_model_blur_robust.keras — on Drive

7.3 BirdLens Web Interface

A complete web interface named BirdLens was deployed this week. The interface is accessible via a public URL generated through Cloudflare Tunnel and requires no installation on the user's device. The system accepts any bird image, runs the full quality check and enhancement pipeline, and returns species predictions with confidence bars.

Week 7 — BirdLens Web Interface Features		
Quality Checker • Laplacian variance blur detection • Brightness analysis • Threshold: blur<60, bright<40 • Warning shown to user	Auto Enhancement • UnsharpMask deblurring • Sharpness boost $\times 2.5$ • Contrast enhancement $\times 1.3$ • Brightness correction $\times 2.2$	TTA + Confidence Tiers • 5-pass prediction averaging • HIGH: $\geq 80\%$ confidence • MEDIUM: 50–80% • UNCERTAIN: $< 50\% \rightarrow$ Top-3

Figure 3: Three core features of the BirdLens interface — quality checker, auto-enhancement, and TTA with confidence tiers

Interface Features

Feature	Implementation
Drag & Drop Upload	Accepts any image format, any quality level
Quality Checker	Laplacian variance (blur<60) + brightness threshold (<40/255)
Auto Enhancement	UnsharpMask + Sharpness $\times 2.5$ + Contrast $\times 1.3$ for blur; Brightness $\times 2.2$ for dark
TTA Prediction	5-pass averaging — original + brightness/contrast/sharpness variants
Confidence Tiers	HIGH $\geq 80\%$ MEDIUM 50–80% UNCERTAIN $< 50\%$ showing Top-3
Animated Results	Confidence bars animate on load, color-coded by tier
Public Access	Cloudflare Tunnel — works on any device, no installation

7.4 Problems & Solutions

Problem	Solution
---------	----------

Blur severe accuracy stuck at 19–21% despite multiple training runs	Switched strategy from training to pre-prediction image enhancement — more effective approach
Gradio dependency conflicts on Python 3.12 prevented web interface deployment	Built custom Flask + HTML interface — no external UI dependencies required
Ngrok required account signup — public URL not available without registration	Switched to Cloudflare Tunnel (cloudflared) — free, no signup, works immediately
Enhancement only improved blur severe by +3.4% in quick test	Image enhancement still a net positive — adds sharpness, contrast, and edge recovery before prediction

7.5 Plan for Week 8

Week 8 will focus on YOLO object detection integration. The YOLOv8 model will be added to the pipeline to detect birds within full-scene images and video streams. Detected bird regions will be automatically cropped and passed to the EfficientNetV2S classifier for species identification. The BirdLens web interface will be updated to support full-scene image upload with detection overlays. The final report structure will also begin during this week.